IOT ENABLED 3-PHASE POWER QUALITY & ENERGY METER

TUE-17

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RESPONSE TO FEEDBACK FROM MILESTONE-1

1. **FEEDBACK**: Need to calculate the amount of data stored in RPI in case of a internet outage

   RESPONSE: Around 7.5kb of data will be stored

2. **FEEDBACK:** Addition of novelty in this version compared to the previous year's project

   RESPONSE:

   a) Addition of relay for remote control of loads

   b) Use of server for storing and fetching data

   c) Emergency battery backup for RPI

PROJECT MANAGEMENT PLAN

Things to be done for Milestone 2

1. Server backend and database connection
2. UI design and setup
3. HTTP request setup
4. Authentication setup and home screen (application)
5. Learning about the evaluation board
6. PCB circuit and schematic design
7. HTTP and SQLite setup on RPI

PROJECT MANAGEMENT PLAN

Things we did for Milestone 2

1. Server backend and database connection
2. UI design and setup
3. HTTP request setup
4. Authentication setup and home screen (application)
5. Learning about the evaluation board
6. PCB circuit and schematic design
7. Integration between the Evaluation board and RPI
8. Completed Server setup
9. Tested Evaluation board with 220 V supply
10. Established SPI Communication between RPI and the Evaluation Board
11. Completed PCB layout
12. API authentication

DEVIATIONS FROM ORIGINAL PLAN

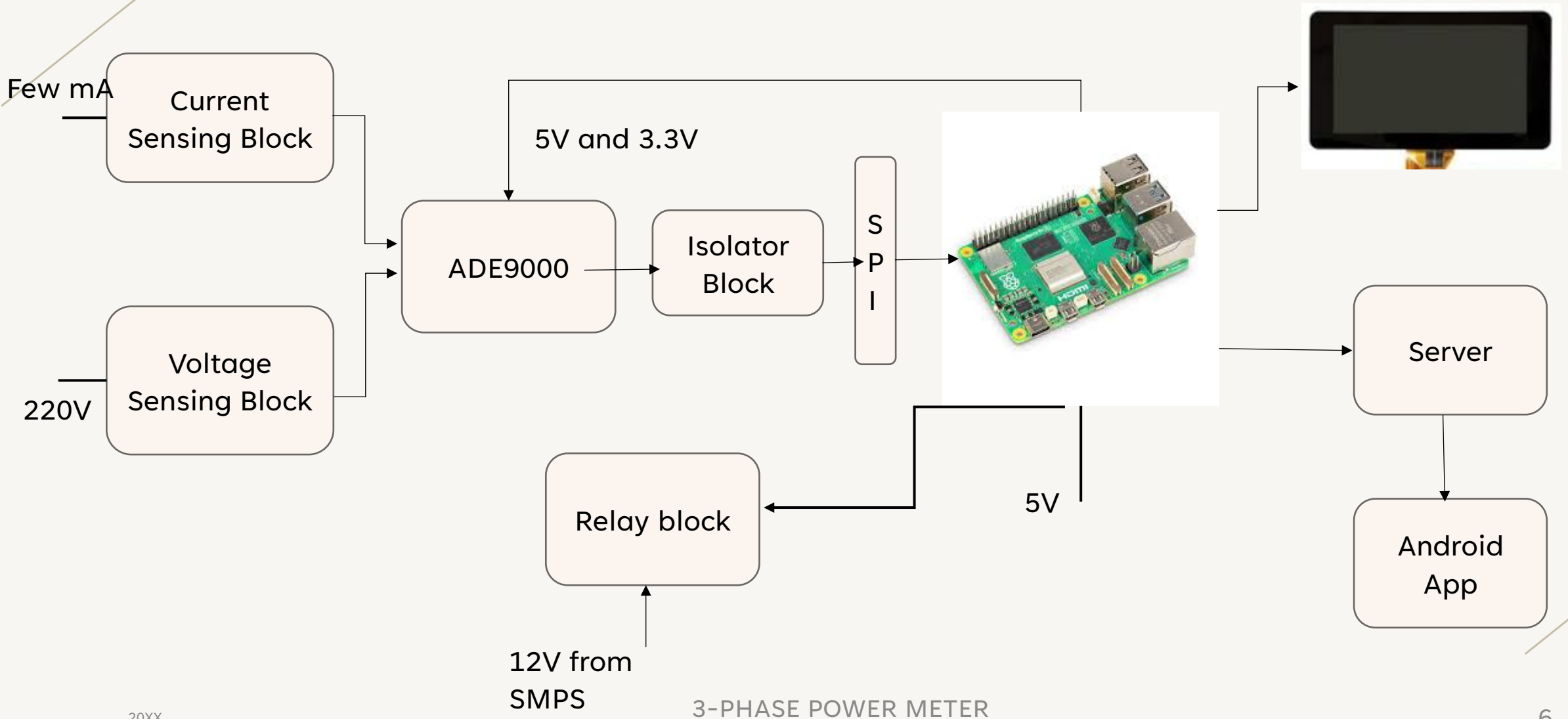
Extra tasks completed:

1. Integration between the Evaluation board and RPI
2. Completed Server setup
3. Tested Evaluation board with 220 V supply
4. Established SPI Communication between RPI and the Evaluation Board
5. Completed PCB layout

Incomplete tasks:

1. HTTP and SQLite setup on RPI

BLOCK DIAGRAM



PRINCIPLE OF OPERATION

1. Voltage and current values are properly down-converted as per the desired range of ADE.
2. ADUM6404 and ADUM4151 isolater ICs are used to isolate the ADE high-voltage side from the low-voltage side.
3. Voltage level shifters ICs are used wherever required to ensure proper voltage levels.
4. ADE9000 board will get powered using 5V and 3.3V from RPI.
5. Using the header interface, 5 SPI connections and a few interrupt signals are transferred from the ADE board to the RPI
6. Some critical data like energy is sent to the server. So that on system reset older data can be fetched.
7. Some critical data will be stored locally when device is offline.

TESTING PLANS

1. Preliminary Testing:
   (a) Testing of the ADE9000 evaluation board using RPI (done)
   (b) Testing of touch display (done)
   (c) Testing of final PCB using RPI
2. Intermediate Testing:
   (a) Testing of server and application integration
   (b) Testing of server and RPI integration
   (c) PCB soldering and testing
3. Final Testing:
   (a) Testing with all subsystems integrated

PRELIMINARY TESTING RESULTS

1. ADE9000 Evaluation Board tested with bulb connected on 1-phase
2. Established SPI protocol between Evaluation board and RPI with and without load
3. Made preliminary UI for display and tested it with dummy data.

Issues: No deviations from the expected result were found as of now.

TOWARDS MILESTONE-3

Projected tasks for Milestone 3-

1. HTTP and SQLite setup on RPI
2. UI design for RPI and application
3. Run all tasks together on RPI
4. Integrating server with RPI and its debugging
5. Soldering Components on PCB
6. Testing and debugging code with 3-phase load

THANK YOU